

MACMILLAN PASS, YT, Canada

WMO: 719900

Lat: 63.244N

Lon: 130.037W

Elev: 4524

StdP: 12.45

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-33.5	-29.3	-41.0	0.6	-33.5	-36.6	0.8	-29.2	25.7	26.0	21.7	19.3	5.0	30	0.854

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	69.3	52.2	65.3	50.3	61.9	49.1	53.5	66.0	51.9	62.3	50.4	59.6	7.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
48.6	59.9	54.2	47.1	56.6	53.2	45.6	53.5	52.3	23.9	66.0	22.9	62.5	22.0	59.9	59.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.9	16.7	14.3	-37.7	74.0	4.9	3.9	-41.2	76.8	-44.0	79.1	-46.7	81.3	-50.2	84.2		
(5)			-37.6	55.7	5.1	1.5	-41.3	56.7	-44.2	57.6	-47.1	58.4	-50.8	59.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	22.5	1.0	0.2	2.3	18.0	35.1	48.0	51.1	47.1	36.5	21.6	5.6	1.8	(6)
(7)		DBStd	22.26	16.17	14.37	14.71	10.35	8.67	5.53	4.53	6.02	6.35	9.21	13.23	12.90	(7)
(8)		HDD50	10200	1518	1393	1478	961	464	100	40	131	407	882	1333	1493	(8)
(9)		HDD65	15513	1983	1813	1943	1411	926	509	430	554	856	1347	1783	1958	(9)
(10)		CDD50	160	0	0	0	0	3	41	74	41	1	0	0	0	(10)
(11)		CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH74	24	0	0	0	0	0	12	5	7	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	5.8	6.0	5.7	6.1	5.6	5.9	6.1	6.0	5.9	6.0	5.5	5.3	5.0	(14)
(15)	Precipitation	PrecAvg	17.1	1.1	0.9	0.8	0.6	1.1	2.0	2.5	2.2	1.9	1.5	1.3	1.1	(15)
(16)		PrecMax	21.5	2.0	1.7	1.9	1.7	1.8	5.2	6.0	3.7	3.1	2.6	3.4	2.4	(16)
(17)		PrecMin	12.0	0.4	0.2	0.2	0.1	0.4	1.1	1.1	0.9	0.3	0.7	0.5	0.4	(17)
(18)		PrecStd	2.0	0.4	0.4	0.4	0.4	0.3	0.7	1.0	0.8	0.5	0.4	0.6	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.3	31.6	37.4	40.3	64.0	75.9	73.1	73.7	58.6	45.1	35.7	31.6	(19)
(20)			MCWB	32.4	28.9	32.3	33.4	45.9	53.6	53.7	54.2	46.4	39.8	32.4	29.4	(20)
(21)		2%	DB	30.5	27.1	30.0	36.8	56.9	68.6	68.1	68.1	52.8	39.0	30.9	27.7	(21)
(22)			MCWB	28.6	25.5	27.0	31.3	43.0	51.4	51.7	52.6	42.9	35.3	28.7	25.9	(22)
(23)		5%	DB	26.8	23.9	25.8	34.1	51.2	63.6	64.8	63.3	49.2	35.7	26.6	24.0	(23)
(24)			MCWB	25.0	22.4	23.9	29.7	40.5	48.5	50.8	50.4	41.3	33.1	24.9	22.4	(24)
(25)		10%	DB	23.0	19.6	21.7	31.4	47.2	59.8	61.7	58.9	46.1	32.9	23.7	19.7	(25)
(26)			MCWB	21.5	18.3	19.9	28.0	38.4	46.9	49.9	48.5	39.8	30.6	22.4	18.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.0	29.2	33.0	34.6	46.5	55.3	55.0	55.4	48.2	40.2	32.1	29.4	(27)
(28)			MCD B	36.0	31.4	37.4	38.9	61.6	72.1	67.9	69.1	55.2	44.1	34.2	30.9	(28)
(29)		2%	WB	28.7	25.8	27.7	31.8	43.8	52.4	53.3	53.4	44.3	35.9	28.4	26.2	(29)
(30)			MCD B	31.0	27.2	30.1	35.2	55.7	66.2	64.2	66.3	51.1	38.2	30.3	27.9	(30)
(31)		5%	WB	25.1	22.2	24.2	30.1	41.2	50.1	51.9	51.4	42.1	33.0	24.5	22.1	(31)
(32)			MCD B	26.6	23.7	26.0	33.2	50.5	61.3	61.4	61.5	47.8	35.3	25.9	23.6	(32)
(33)		10%	WB	21.6	17.9	20.3	28.2	38.9	48.1	50.7	49.4	40.3	30.8	21.7	18.2	(33)
(34)			MCD B	23.0	19.0	22.1	31.3	46.7	57.2	59.5	57.7	45.2	32.7	22.9	19.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.5	12.5	13.6	14.8	15.4	19.4	18.5	17.4	13.7	11.8	11.5	11.9	(35)
(36)			MCDBR	12.9	13.4	12.9	14.6	20.2	25.6	24.7	25.2	19.4	12.5	12.0	15.2	(36)
(37)			MCWBR	11.3	12.6	10.7	10.4	10.5	11.9	11.4	12.1	11.4	9.2	10.8	14.7	(37)
(38)		5% WB	MCD BR	13.1	13.4	12.5	12.8	19.8	23.1	20.8	23.6	17.0	11.6	12.6	15.4	(38)
(39)			MCWBR	11.8	12.4	10.5	9.6	10.4	11.1	10.3	11.5	10.5	8.8	11.5	14.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.160	0.218	0.237	0.287	0.310	0.327	0.341	0.318	0.274	0.230	0.161	0.151	(40)	
(41)		taud	2.041	2.264	2.278	2.161	2.230	2.405	2.386	2.484	2.534	2.319	2.005	1.920	(41)	
(42)		Ebn at Noon	174	245	282	284	286	283	275	274	268	234	162	93	(42)	
(43)		Edh at Noon	9	17	26	36	38	33	32	27	21	16	9	4	(43)	
(44)	All-Sky Solar Radiation	RadAvg	66	298	874	1569	1806	1666	1489	1162	711	300	95	30	(44)	
(45)		RadStd	8	37	69	82	110	92	91	109	82	40	13	2	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page